

Objective

The objective of the Data Pre-processing phase is to prepare the loan prediction dataset for machine learning by improving data quality and ensuring consistency. This phase includes handling missing values, encoding categorical variables, balancing the dataset, scaling numerical features, and splitting the dataset into training and testing sets. Proper preprocessing helps improve model accuracy, reliability, and overall performance.

Description

Data preprocessing is one of the most important stages in the machine learning workflow. Raw datasets often contain missing values, categorical features, imbalanced classes, and inconsistencies that can negatively affect model performance. During this phase, several preprocessing techniques are applied to clean and transform the data into a suitable format for machine learning algorithms.

The preprocessing phase consists of the following stories:

Story 1: Checking and Handling Values

The dataset is examined to identify missing values, duplicate records, and inconsistent data using functions such as `df.shape`, `df.info()`, and `df.isnull().sum()`. Missing numerical values are replaced with the mean of their respective columns, while missing categorical values are replaced with the mode (most frequent value). Categorical features such as Gender, Married, Dependents, Education, Self_Employed, Credit_History, and Property_Area are encoded into numerical values to ensure compatibility with machine learning algorithms.

Story 2: Balancing the Dataset

The loan approval dataset may contain an unequal number of approved and rejected loan applications, leading to class imbalance. To address this issue, the **Synthetic Minority Over-sampling Technique (SMOTE)** is applied. SMOTE generates synthetic samples for the minority class using the K-Nearest Neighbors (KNN) algorithm, resulting in a balanced dataset that improves model fairness and predictive performance.

Story 3: Scaling the Data

Feature scaling is performed to ensure that numerical features have a consistent scale. Since algorithms such as K-Nearest Neighbors and Logistic Regression rely on distance calculations, scaling prevents features with larger values from dominating the learning process. The **StandardScaler** method from Scikit-learn is applied to the input features, transforming the data into a standardized format while leaving the target variable unchanged.

Story 4: Splitting Data into Training and Test Sets

After preprocessing, the dataset is divided into input features (X) and the target variable (y). The `train_test_split()` function from Scikit-learn is used to split the dataset into training and testing subsets. Typically, 80% of the data is used for training and 20% is reserved for testing. A fixed `random_state` value is used to ensure reproducibility of the results.

Tools and Libraries Used

- Python
 - Pandas
 - NumPy
 - Scikit-learn
 - Imbalanced-learn (SMOTE)
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Expected Outcome

After completing the Data Pre-processing phase:

- Missing values are identified and handled.
- Categorical variables are converted into numerical values.
- Duplicate and inconsistent records are removed or corrected.
- The dataset is balanced using SMOTE.
- Numerical features are standardized through feature scaling.
- The dataset is successfully split into training and testing sets.

- The processed dataset is ready for machine learning model training.

Conclusion

The Data Pre-processing phase transforms the raw loan prediction dataset into a clean, structured, and machine learning-ready format. By handling missing values, encoding categorical features, balancing the target classes, scaling numerical attributes, and splitting the dataset into training and testing sets, the quality of the data is significantly improved. These preprocessing steps provide a strong foundation for building accurate and reliable machine learning models in the Smart Lender application.